

- GET message becoming the payload of a TCP segment. The TCP segment is placed in a datagram and sent and delivered to `www.google.com` as in steps 18–20 above.
23. The HTTP server at `www.google.com` reads the HTTP GET message from the TCP socket, creates an **HTTP response** message (Section 2.2), places the requested Web page content in the body of the HTTP response message, and sends the message into the TCP socket.
  24. The datagram containing the HTTP reply message is forwarded through the Google, Comcast, and school networks, and arrives at Bob's laptop. Bob's Web browser program reads the HTTP response from the socket, extracts the html for the Web page from the body of the HTTP response, and finally (*finally!*) displays the Web page!

Our scenario above has covered a lot of networking ground! If you've understood most or all of the above example, then you've also covered a lot of ground since you first read Section 1.1, where we wrote “much of this book is concerned with computer network protocols” and you may have wondered what a protocol actually was! As detailed as the above example might seem, we've omitted a number of possible additional protocols (e.g., NAT running in the school's gateway router, wireless access to the school's network, security protocols for accessing the school network or encrypting segments or datagrams, network management protocols), and considerations (Web caching, the DNS hierarchy) that one would encounter in the public Internet. We'll cover a number of these topics and more in the second part of this book.

Lastly, we note that our example above was an integrated and holistic, but also very “nuts and bolts,” view of many of the protocols that we've studied in the first part of this book. The example focused more on the “how” than the “why.” For a broader, more reflective view on the design of network protocols in general, see [Clark 1988, RFC 5218].

## 5.8 Summary

In this chapter, we've examined the link layer—its services, the principles underlying its operation, and a number of important specific protocols that use these principles in implementing link-layer services.

We saw that the basic service of the link layer is to move a network-layer datagram from one node (host, switch, router, WiFi access point) to an adjacent node. We saw that all link-layer protocols operate by encapsulating a network-layer datagram within a link-layer frame before transmitting the frame over the link to the adjacent node. Beyond this common framing function, however, we learned that different link-layer protocols provide very different link access, delivery, and transmission services. These differences are due in part to the wide variety of link types over which link-layer protocols must operate. A simple point-to-point link has a single sender and receiver communicating over a single “wire.” A multiple access link is shared among many senders and receivers; consequently, the link-layer protocol for a multiple access channel has a protocol (its multiple access protocol) for coordinating link access. In the case of MPLS, the “link” connecting two adjacent nodes (for

example, two IP routers that are adjacent in an IP sense—that they are next-hop IP routers toward some destination) may actually be a *network* in and of itself. In one sense, the idea of a network being considered as a link should not seem odd. A telephone link connecting a home modem/computer to a remote modem/router, for example, is actually a path through a sophisticated and complex telephone *network*.

Among the principles underlying link-layer communication, we examined error-detection and -correction techniques, multiple access protocols, link-layer addressing, virtualization (VLANs), and the construction of extended switched LANs and data center networks. Much of the focus today at the link layer is on these switched networks. In the case of error detection/correction, we examined how it is possible to add additional bits to a frame's header in order to detect, and in some cases correct, bit-flip errors that might occur when the frame is transmitted over the link. We covered simple parity and checksumming schemes, as well as the more robust cyclic redundancy check. We then moved on to the topic of multiple access protocols. We identified and studied three broad approaches for coordinating access to a broadcast channel: channel partitioning approaches (TDM, FDM), random access approaches (the ALOHA protocols and CSMA protocols), and taking-turns approaches (polling and token passing). We studied the cable access network and found that it uses many of these multiple access methods. We saw that a consequence of having multiple nodes share a single broadcast channel was the need to provide node addresses at the link layer. We learned that link-layer addresses were quite different from network-layer addresses and that, in the case of the Internet, a special protocol (ARP—the Address Resolution Protocol) is used to translate between these two forms of addressing and studied the hugely successful Ethernet protocol in detail. We then examined how nodes sharing a broadcast channel form a LAN and how multiple LANs can be connected together to form larger LANs—all *without* the intervention of network-layer routing to interconnect these local nodes. We also learned how multiple virtual LANs can be created on a single physical LAN infrastructure.

We ended our study of the link layer by focusing on how MPLS networks provide link-layer services when they interconnect IP routers and an overview of the network designs for today's massive data centers. We wrapped up this chapter (and indeed the first five chapters) by identifying the many protocols that are needed to fetch a simple Web page. Having covered the link layer, *our journey down the protocol stack is now over!* Certainly, the physical layer lies below the link layer, but the details of the physical layer are probably best left for another course (for example, in communication theory, rather than computer networking). We have, however, touched upon several aspects of the physical layer in this chapter and in Chapter 1 (our discussion of physical media in Section 1.2). We'll consider the physical layer again when we study wireless link characteristics in the next chapter.

Although our journey down the protocol stack is over, our study of computer networking is not yet at an end. In the following four chapters we cover wireless networking, multimedia networking, network security, and network management. These four topics do not fit conveniently into any one layer; indeed, each topic crosscuts many layers. Understanding these topics (billed as advanced topics in

some networking texts) thus requires a firm foundation in all layers of the protocol stack—a foundation that our study of the link layer has now completed!



## Homework Problems and Questions

### Chapter 5 Review Questions

#### SECTIONS 5.1–5.2

- R1. Consider the transportation analogy in Section 5.1.1. If the passenger is analogous to a datagram, what is analogous to the link layer frame?
- R2. If all the links in the Internet were to provide reliable delivery service, would the TCP reliable delivery service be redundant? Why or why not?
- R3. What are some of the possible services that a link-layer protocol can offer to the network layer? Which of these link-layer services have corresponding services in IP? In TCP?

#### SECTION 5.3

- R4. Suppose two nodes start to transmit at the same time a packet of length  $L$  over a broadcast channel of rate  $R$ . Denote the propagation delay between the two nodes as  $d_{\text{prop}}$ . Will there be a collision if  $d_{\text{prop}} < L/R$ ? Why or why not?
- R5. In Section 5.3, we listed four desirable characteristics of a broadcast channel. Which of these characteristics does slotted ALOHA have? Which of these characteristics does token passing have?
- R6. In CSMA/CD, after the fifth collision, what is the probability that a node chooses  $K = 4$ ? The result  $K = 4$  corresponds to a delay of how many seconds on a 10 Mbps Ethernet?
- R7. Describe polling and token-passing protocols using the analogy of cocktail party interactions.
- R8. Why would the token-ring protocol be inefficient if a LAN had a very large perimeter?

#### SECTION 5.4

- R9. How big is the MAC address space? The IPv4 address space? The IPv6 address space?
- R10. Suppose nodes A, B, and C each attach to the same broadcast LAN (through their adapters). If A sends thousands of IP datagrams to B with each encapsulating frame addressed to the MAC address of B, will C's adapter process these frames? If so, will C's adapter pass the IP datagrams in these frames to the network layer C? How would your answers change if A sends frames with the MAC broadcast address?